Eduardo Fernandes Montesuma

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Expertise Optimal Transport, Domain Adaptation, Transfer Learning, Foundation Models

Education

Université Paris-Saclay, Paris, Île-de-France. France — October 2021 – October 2024
PhD in Mathematical Computer Science.
Thesis Subject: Multi-Source Domain Adaptation through Wasserstein Barycenters
Thesis Director: Antoine Souloumiac, PhD
Advisor: Fred Maurice Ngolè Mboula, PhD

INSA de Rennes, Rennes, Bretagne. France — September 2018 – August 2020
Master's Degree in Electronic Engineering and Industrial Informatics — GPA 14.80/20.00
Master Thesis: Optimal Transport for Domain Adaptation with Applications to Bacteria Stain Classification
Advisor: Fred-Maurice Ngolè Mboula, PhD

Universidade Federal do Ceará, Fortaleza, Ceará. Brazil — January 2015 – April 2021
Bachelor's Degree in Computer Engineering — GPA 8.94/10.00
Bachelor Thesis: Cross-Domain Fault Diagnosis through Optimal Transport
Advisor: Michela Mulas, PhD

Research Experience

Sigma Nova, Paris, France.
AI Researcher — *August 2025 – Present*
As an AI Researcher at Sigma Nova, I am working with pre-training, fine-tuning and transfer learning of large scale foundation models for neuroscience and time series forecasting.
Associated Publications: [J1, C1, C2, C5, C6, P2, P3]

GetVocal, Paris, France.
Applied AI Researcher — *October 2024 – July 2025*
As an applied AI researcher at GetVocal, I am in charge of R&D of conversational agents, especially on topics such as automatic speech recognition, large language models and speech synthesis.

Commissariat à l'Énergie Atomique et aux Énergies Alternatives, Saclay, France.
PhD Student — *October 2021 – October 2024*
As a PhD Student, I am in charge of studying the contributions of Optimal Transport for Machine Learning, Transfer Learning and Domain Adaptation.
Associated Publications: [J2, J3, J4, C3, C4, C7, C8, C9, C10, C11, C12, P1, P4, S1, T1]

Dell Lead, Fortaleza, Brazil.
Scientific Researcher in Data Mining — *April 2021 – September 2021*
As a scientific researcher I was in charge of **mining textual data**, **building visualizations** and exploring strategies for **machine reading comprehension**.
Associated Publications: [C15]

Commissariat à l'Énergie Atomique et aux Énergies Alternatives, Saclay, France.
Research Intern in Artificial Intelligence — *March 2020 – August 2020*
I was in charge of implementing **Optimal Transport for Domain Adaptation** methods for a problem involving transfer learning on the classification of bacteria cell stain.
Associated Publications: [C13, C14]

Institut d'Électronique et Télécommunications de Rennes, Rennes, France.
Research Intern in Deep Learning for Image Restoration *May 2019 – September 2019*
I was responsible for implementing a software for comparing **image restoration methods**, with a focus on deep neural network methods. Our solution is open-source and hosted on Github.
Associated Publications: [C16]

Universidade Federal do Ceará, Fortaleza, Brazil.
Research Intern in Pattern Recognition *August 2016 – August 2017*
As an intern in the Computer Vision for Detection and Identification of Cervix Diseases (SISCOLO) project, I was in charge of studying and implementing classifiers for identifying cancer on cervix cells.
Associated Publications: [C17]

Teaching Experience

École Polytechnique, Palaiseau, France.
Python for Data Science *September 2023 - January 2024*
Over the first term of the academic year 2023 – 2024, I am in charge of an introductory course on Python for Data Science for students of the masters in Applied Maths and Statistics at École Polytechnique.

Universidade Federal do Ceará, Fortaleza, Brazil.
An Introduction to Control Theory *January 2018 – August 2018*
Over one semester, I was in charge with preparing problem solution lectures, as well as lectures on special topics on control systems (e.g., Model Predictive Control).

Scholarships and Awards

BRAFITEC/France program, granted by Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES).

Programming

Python (Tensorflow/Keras, Pytorch, Scikit-Learn, PythonOT), Matlab (simulink).

Languages

Portuguese (mother tongue), French (B2), English (B2), Spanish (A2).

Journal Articles

[J1] Eufrazio, Rafael Pereira, **Eduardo Fernandes Montesuma**, and Charles Casimiro Cavalcante. "Nonlinear dimensionality reduction through optimal transport between incomparable spaces." Information Geometry, 1-36, 2026.

[J2] **Fernandes Montesuma, Eduardo**, A. E. Habazi, and F. N. Mboula. "Unsupervised Anomaly Detection through Mass Repulsing Optimal Transport." Transactions on Machine Learning Research, 2025.

[J3] **Fernandes Montesuma, Eduardo**, Fred Maurice Ngolè Mboula, and Antoine Souloumiac. "Optimal Transport for Domain Adaptation through Gaussian Mixture Models." Transactions on Machine Learning Research, 2025.

[J4] **Montesuma, Eduardo Fernandes**, Fred Ngolè Mboula, and Antoine Souloumiac. "Recent advances in optimal transport for machine learning." IEEE Transactions on Pattern Analysis and Machine Intelligence, 47(2), 1161-1180, 2024.

Conference and Workshop Papers

[C1] Bendou, Y., Ezzahir, O., **Montesuma, E. F.**[1], Mahuas, G., Shevchenko, V., and Gartrell, M. "ReBaPL: Repulsive Bayesian Prompt Learning." Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026.

[1] Equal contribution, corresponding author

[C2] **Fernandes Montesuma, Eduardo**, Yassir Bendou, and Mike Gartrell. "Wasserstein Gradient Flows for Scalable and Regularized Barycenter Computation." Proceedings of the 42nd Conference on Uncertainty in Artificial Intelligence (UAI), Proceedings of Machine Learning Research 337, 4595-4606. PMLR, 2026.

[C3] **Montesuma, Eduardo Fernandes**, M. Mulas, Fred Ngolè Mboula, F. Corona, and Antoine Souloumiac. "Benchmarking Domain Adaptation for Chemical Processes on the Tennessee Eastman Process." Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ML4CCE Workshop at ECML-PKDD 2024). Cham: Springer Nature Switzerland, 2026.

[C4] Clain, R., **Fernandes Montesuma, Eduardo**, and F. N. Mboula. "Decentralized Federated Dataset Dictionary Learning for Multi-Source Domain Adaptation." 2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP). IEEE, 2025.

[C5] **Montesuma, Eduardo Fernandes**. "KD^{2}M: A Unifying Framework for Feature Knowledge Distillation." International Conference on Geometric Science of Information (GSI), 121-130. Cham: Springer Nature Switzerland, 2025.

[C6] Eufrazio, Rafael Pereira, **Eduardo Fernandes Montesuma**, and Charles Casimiro Cavalcante. "A dimensionality reduction technique based on the Gromov-Wasserstein distance." International Conference on Geometric Science of Information (GSI), 111-120. Cham: Springer Nature Switzerland, 2025.

[C7] **Montesuma, Eduardo Fernandes**, Fred Ngolè Mboula, and Antoine Souloumiac. "Lighter, better, faster multi-source domain adaptation with gaussian mixture models and optimal transport." Joint European Conference on Machine Learning and Knowledge Discovery in Databases (ECML-PKDD). Cham: Springer Nature Switzerland, 2024.

[C8] Espinoza Castellon, F., **Fernandes Montesuma, E.**, Mboula, F. N., Mayoue, A., Souloumiac, A., & Gouy-Pallier, C. "Federated dataset dictionary learning for multi-source domain adaptation." 2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP). IEEE, 2024.

[C9] **Fernandes Montesuma, Eduardo**, Mboula, F. N., & Souloumiac, A. "Multi-source domain adaptation meets dataset distillation through dataset dictionary learning." 2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP). IEEE, 2024.

[C10] **Montesuma, Eduardo Fernandes**, S. Le Stanc, and Fred Ngolè Mboula. "Online Multi-Source Domain Adaptation Through Gaussian Mixtures and Dataset Dictionary Learning." 2024 IEEE 34th International Workshop on Machine Learning for Signal Processing (MLSP). IEEE, 2024.

[C11] **Montesuma, Eduardo Fernandes**, Fred Ngolè Mboula, and Antoine Souloumiac. "Multi-Source Domain Adaptation Through Dataset Dictionary Learning in Wasserstein Space." 26th European Conference on Artificial Intelligence (ECAI), 372, 1739-1746. 2023.

[C12] **Fernandes Montesuma, Eduardo**, Mulas, M., Corona, F., and Mboula, F. M. N. "Cross-domain fault diagnosis through optimal transport for a cstr process." IFAC-PapersOnLine, 55(7), 946-951. 2022.

[C13] **Montesuma, Eduardo Fernandes**, and Fred Maurice Ngole Mboula. "Wasserstein Barycenter for Multi-Source Domain Adaptation." Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 16785-16793. 2021.

[C14] **Montesuma, Eduardo Fernandes**, and Fred-Maurice Ngolè Mboula. "Wasserstein Barycen-

ter Transport for Acoustic Adaptation." 2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP). IEEE, 2021.

[C15] **Fernandes Montesuma, Eduardo**, et al. "An Empirical Study of Information Retrieval and Machine Reading Comprehension Algorithms for an Online Education Platform." Anais do XIII Simpósio Brasileiro de Tecnologia da Informação e da Linguagem Humana (STIL). SBC, 2021.

[C16] Lemarchand, Florian, **Fernandes Montesuma, Eduardo**, M. Pelcat, and E. Nogues. "Opendenoising: an extensible benchmark for building comparative studies of image denoisers." 2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP). IEEE, 2020.

[C17] **Montesuma, Eduardo Fernandes**, Alencar, Levi P. and Barreto, Guilherme A. "Avaliação de Algoritmos de Classificação de Padrões na Detecção de Câncer do Colo do Útero." VIII Simpósio de Instrumentação e Imagens Médicas (SIIM), 2017.

Preprints and Works Under Review

[P1] Clain, R., **Fernandes Montesuma, Eduardo**, and F. N. Mboula. "DeFed-GMM-DaDiL: A Decentralized Federated Framework for Domain Adaptation." arXiv preprint arXiv:2605.04324, 2026.

[P2] Eufrazio, Rafael Pereira, **Eduardo Fernandes Montesuma**, and Charles Casimiro Cavalcante. "Structure-Preserving Multi-View Embedding Using Gromov-Wasserstein Optimal Transport." arXiv preprint arXiv:2604.02610, 2026.

[P3] Eufrazio, Rafael Pereira, **Eduardo Fernandes Montesuma**, and Charles Casimiro Cavalcante. "Gromov-Wasserstein Methods for Multi-View Relational Embedding and Clustering." arXiv preprint arXiv:2604.23912, 2026.

[P4] **Fernandes Montesuma, Eduardo**, F. E. Castellon, F. N. Mboula, A. Mayoue, A. Souloumiac, and C. Gouy-Pallier. "Dataset dictionary learning in a wasserstein space for federated domain adaptation." arXiv preprint arXiv:2407.11647, 2024.

</> Software

[S1] Flamary, R., Vincent-Cuaz, C., Courty, N., Gramfort, A., Kachaiev, O., Tran, H. Q., **Fernandes Montesuma, E.**, et al. "POT: Python Optimal Transport (version 0.9.5)." `https://github.com/PythonOT/POT`, 2024.

Theses

[T1] **Montesuma, Eduardo Fernandes**. "Multi-Source Domain Adaptation through Wasserstein Barycenters." PhD Thesis, Université Paris-Saclay, 2024.

[T2] **Montesuma, Eduardo Fernandes**. "Cross-Domain Fault Diagnosis through Optimal Transport." Bachelor Thesis, Universidade Federal do Ceará, 2021.